

Comparing Exploration Algorithms in the 3D Maze

A Course-Project Experiment Design + Coverage-Heatmap Evolution

Challenging Problems in Reinforcement Learning (SS 2026)

Topic: Exploration vs. Exploitation

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Abstract

We compare five exploration mechanisms—Learning Progress Monitoring (LPM), Random Network Distillation (RND), the Intrinsic Curiosity Module (ICM), a raw next-state prediction-error bonus (MSE), and a no-intrinsic A2C baseline (NONE)—in the Miniworld 3D maze of Hou, An & Du (2026), *Beyond Noisy-TVs: Noise-Robust Exploration via Learning Progress Monitoring*. The maze emits *no extrinsic reward*, so a method’s quality is its *spatial coverage*, and the noisy-TV failure mode is geometric: a random-RGB “noise wall” at $z \approx 8$ separates the two lower spawn rooms from the far room (room4), so a prediction-error agent fascinated by the wall never crosses it. Two questions drive the design: **(RQ1)** how much coverage—and in particular coverage *past the wall*—does each method achieve under the clean vs. noisy variants, and **(RQ2)** how does the *spatial distribution* of exploration *evolve over training time*? RQ2 is answered by a new statistic we add to the LPM codebase: a coverage-heatmap-evolution figure. All methods share one A2C base so that observed differences are attributable to the exploration mechanism rather than the base learner. In a 30-run downscaled study we find one clean reproduction—under action-triggered noise, the raw-pixel method fixates on the noisy TV 99% of the time while LPM avoids it most (2%)—alongside otherwise variance-dominated coverage at this budget; the heatmap-evolution statistic renders the spatial exploration dynamics clearly.

1 Motivation and research questions

The course topic is the exploration–exploitation trade-off. Deep RL drives directed exploration with intrinsic-motivation bonuses (prediction error, novelty, learning progress). The *noisy-TV* problem is the canonical failure mode of prediction-error exploration: an unlearnable source of randomness produces perpetually high prediction error, so a curiosity-driven agent fixates on it. LPM rewards the *improvement* of its dynamics model rather than the error level, and is therefore noise-robust by construction. The 3D maze makes this failure spatial and directly visualisable: the noise wall is a physical place the agent can get stuck at.

The maze geometry is a hand-designed four-room layout in an 18×12 world (continuous units): two spawn rooms below the wall (room1 $x \in [0, 4]$, room2 $x \in [14, 18]$, both $z \in [0, 8]$), the thin grass-textured *noise wall* (room3, $z \in [8, 8.1]$), and the large far room (room4, $z \in [8.1, 12]$). The agent spawns in room1 and must traverse the wall to explore room4.

RQ1 (coverage & noise robustness, reproduction).

Under the noisy variants (noisy_tv: random-RGB wall + 25% sticky actions; action_noise: an action that shows a random CIFAR-10 image) vs. the clean control (nonoise), which methods keep expanding coverage and which stall? How large is LPM’s advantage in coverage *past the wall* over the prediction-error methods (MSE, ICM, RND)?

RQ2 (spatial dynamics, novelty).

How does the spatial distribution of exploration evolve over training? Do prediction-error methods concentrate occupancy at the noise wall over time while LPM’s coverage frontier advances into room4? We answer this with a coverage-heatmap-evolution figure (rows: method; columns: training-time window).

2 Methods

All five methods use the same A2C base (RMSprop, GAE with $\gamma = 0.99$, $\lambda = 0.95$, a single-epoch update every 64 steps, a Nature-style CNN over 160×120 RGB frames). Because the maze extrinsic reward is identically zero, the per-step training reward reduces to a normalised intrinsic term,

$$r_t = \lambda_i \hat{r}_t^i, \quad \hat{r}_t^i = \frac{r_t^i - \mu_t}{\sigma_t + \epsilon}, \quad (1)$$

with $\lambda_i = 0.1$ and (μ_t, σ_t) a running mean/std of the raw intrinsic reward. This uniform running-std normalisation (RND-style) replaces the upstream notebooks’ inconsistent, hand-set offsets so that a single λ_i is comparable across methods; for NONE, $r_t \equiv 0$ and the agent is a pure entropy-regularised random walker.

Class	Method	Mechanism (intrinsic reward r^i)
Learning progress	LPM	$\text{clip}(\eta \mathbb{E}[\varepsilon] - \varepsilon_{\text{actual}}, \leq 0.5)$: predicted-minus-actual next-state error (improvement)
Prediction error	RND	error of a predictor vs. a fixed random target network on the next frame
Prediction error (feat.)	ICM	forward-dynamics error in an inverse-dynamics feature space
Prediction error (pixel)	MSE	raw next-state pixel-prediction error (the pure noisy-TV victim)
None (baseline)	NONE	no intrinsic reward; entropy-regularised A2C random walk

Table 1: The five methods. LPM and MSE share the same pixel-decoder predictor, isolating the effect of the learning-progress signal vs. raw error. ICM and RND are canonical implementations added on the shared CNN encoder (the upstream notebooks’ “ICM” was in fact a pixel decoder, and contained no RND).

3 Experimental design

A single manipulation grid: method $\times$ variant $\times$ seed.

Grid. Five methods $\times$ three variants (**nonoise**, **noisy_tv**, **action_noise**) $\times$ two seeds = 30 runs. Each run is a single environment trained for 2×10^4 steps (downscaled from the upstream 5×10^4 for local compute).

Budget. 30 runs $\times$ ~ 4 min (decoder methods) on Apple-Silicon CPU ≈ 1.5 h. Runs use CPU rather than MPS: the decoder methods (LPM, MSE) are *faster* on CPU because `ConvTranspose2d` falls back off the MPS device. The grid is resumable (completed CSVs are skipped).

4 Metrics

Coverage is measured on the maze’s native 72×48 occupancy grid (cells of 0.25 world units), restricted to the *reachable* cells (the union of the four rooms; the empty corridor between the spawn rooms is excluded). Let the final value of a quantity be its mean over the last 10% of logged updates, reported as mean $\pm$ std over seeds.

- **RQ1 headline — beyond-wall coverage:** $C_{\text{bw}} = (\text{visited reachable cells with } z \geq 8.1) / (\text{reachable cells with } z \geq 8.1)$, i.e. the fraction of room4 explored. Prediction: under noise, MSE/ICM/RND collapse while LPM stays high.
- **Total coverage:** $C = (\text{visited reachable cells}) / (\text{reachable cells})$, reported as a learning curve vs. frames.
- **Time-at-wall:** fraction of steps with $z \in [7.5, 8.5]$ —a direct measure of “fascination with the noise.” Prediction: highest for the pixel prediction-error methods under `noisy_tv`.
- **RQ2 headline — coverage-heatmap evolution (new statistic):** for each variant, a rows(method) $\times$ columns(training-window) grid of top-down occupancy heatmaps, in two views: *occupancy density* (log step-count per cell, revealing where the agent dwells) and *coverage frontier* (cells ever visited up to each window, revealing how the explored region grows). The maze rooms and the $z = 8$ wall line are overlaid for orientation.

5 Results (downscaled run)

We ran the full $5 \times 3 \times 2 = 30$ -run grid at 2×10^4 steps on CPU (≈ 30 min wall-clock). The results are *qualitative*: at this budget the between-seed variance is large, so they should be read as trends, not significance tests.

The clean positive: noisy-TV fixation under action_noise. The sharpest result is the textbook noisy-TV trap. In the `action_noise` variant, action 4 replaces the observation with a random CIFAR-10 image. The raw-pixel predictor (MSE) is captured by it almost completely—it selects action 4 on $98.6\% \pm 1.4\%$ of steps and visits only ~ 10 of the ~ 1500 reachable cells—while LPM selects it on just $2.0\% \pm 1.7\%$ of steps, the *lowest* of all methods and far below the 20% uniform rate (Fig. 1). LPM is noise-robust exactly as predicted: it does not reward the unlearnable image because its dynamics model cannot *improve* on it. ICM (15%) and the NONE baseline (7%) sit between; RND is bimodal across seeds ($22\% \pm 27\%$).

Coverage and beyond-the-wall. The overall coverage ranking (Fig. 2) does *not* cleanly reproduce the paper’s “LPM dominates under noise” headline. The no-intrinsic NONE baseline is competitive everywhere, indicating that at 2×10^4 steps with a single environment and entropy coefficient 0.05, exploration is still largely driven by the policy’s own stochasticity rather than the intrinsic signal. LPM is slow to start (its coverage frontier only advances in the later training windows) and ends mid-pack; MSE actually does well in the pixel-wall `noisy_tv` variant (the wall

Method	TV-action share (action_noise)	Coverage (action_noise)
LPM	0.020 ± 0.017	0.231 ± 0.311
NONE	0.072 ± 0.001	0.345 ± 0.043
ICM	0.150 ± 0.078	0.420 ± 0.006
RND	0.220 ± 0.271	0.108 ± 0.135
MSE	0.986 ± 0.014	0.017 ± 0.017

Table 2: Noisy-TV fixation. MSE (raw pixel error) is trapped by the random-image action and barely moves; LPM avoids it most strongly. Mean $\pm$ std over 2 seeds.

noise is a *place* it can move past, not an action that arrests it) yet collapses under **action_noise** as above; RND collapses on the clean **nonoise** variant (one seed never leaves the spawn room). These are budget- and variance-dominated outcomes, not contradictions of the method—see Limitations.

The new statistic works. The coverage-heatmap-evolution figure (Fig. 3) makes the spatial story legible: ICM and MSE spread across and past the wall over successive windows, the NONE walk diffuses outward, and LPM’s occupancy stays sparse until late. It is the most informative single view of *where* and *when* each method explores, and is the intended contribution of this experiment.

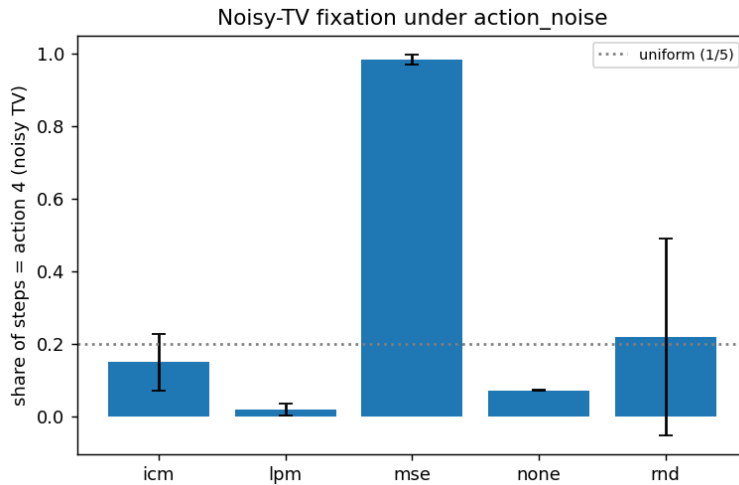


Figure 1: Share of steps spent on the noisy-TV action (action 4) under **action_noise**. MSE fixates ($\sim 99\%$); LPM avoids it most ($\sim 2\%$). Dotted line: uniform $1/5$.

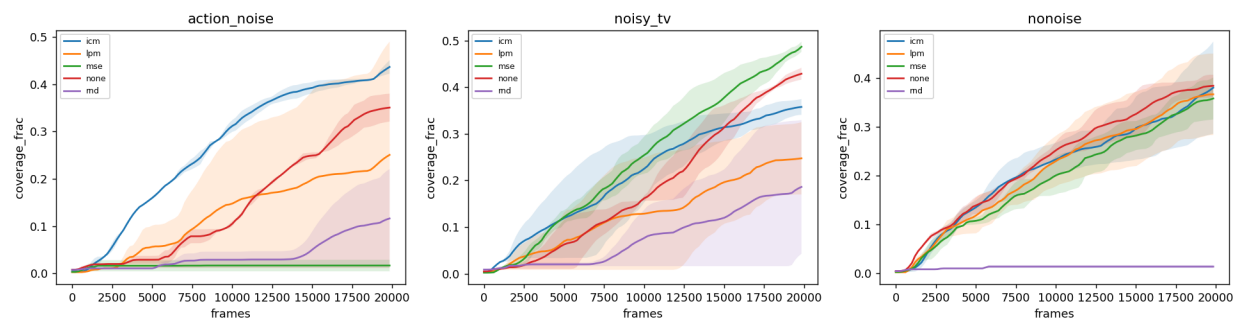


Figure 2: Reachable-cell coverage vs. frames, per variant (mean $\pm$ std over 2 seeds). Note MSE's collapse under **action_noise** (left) and RND's collapse under **nonoise** (right).

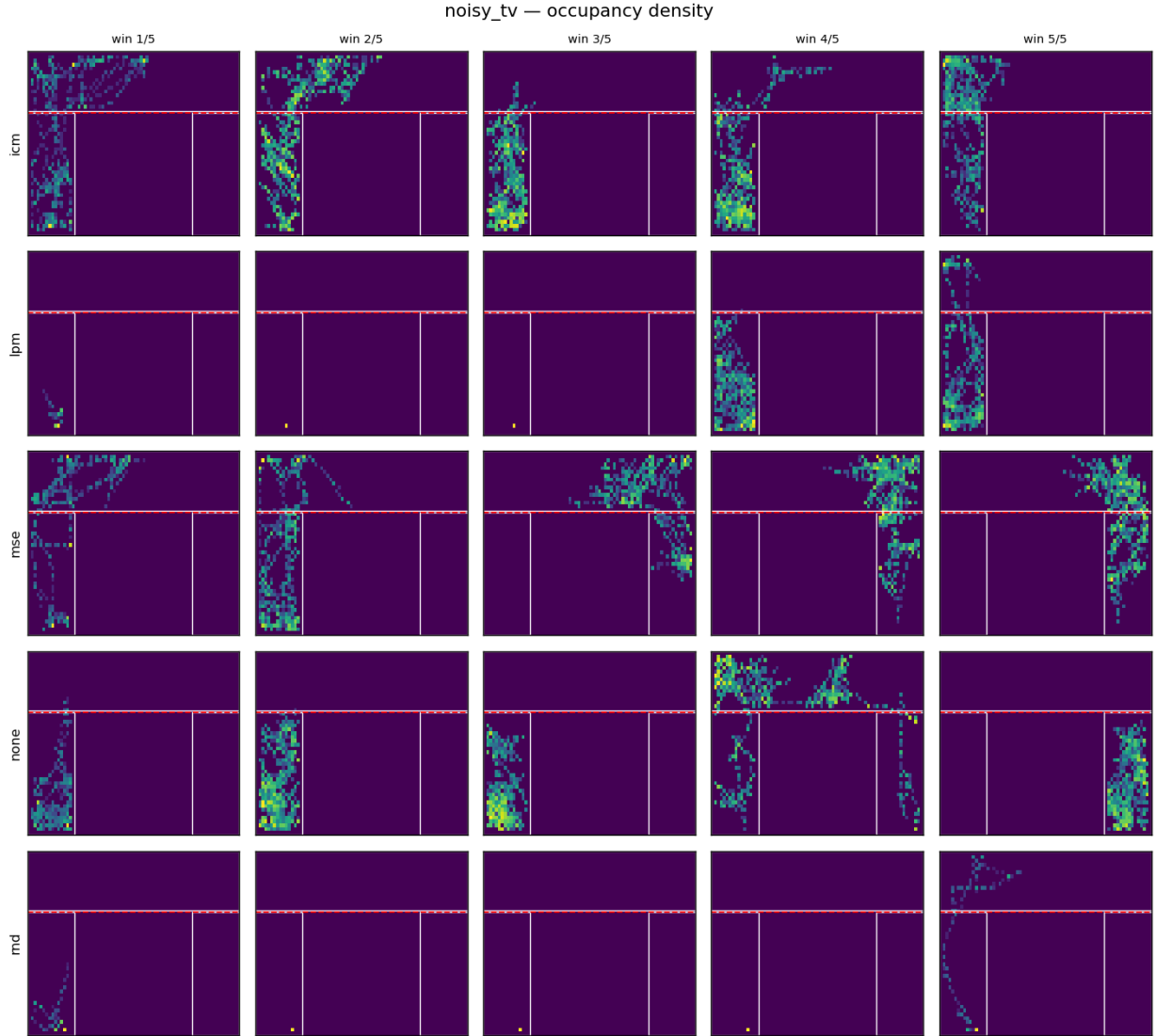


Figure 3: Coverage-heatmap evolution (the new statistic), `noisy_tv` variant. Rows: method; columns: five equal training-time windows. Brightness is log step-count per 0.25-unit cell; the red dashed line is the noise wall ($z = 8$) and the white rectangles are the reachable rooms.

6 Infrastructure and reproducibility

The upstream maze experiment exists only as three Jupyter notebooks with an embedded A2C and no CLI. We extracted the A2C engine, the LPM prediction/uncertainty nets, and the pixel decoder *verbatim* into an importable, CLI-driven package (`LPM_exploration/Miniworld/experiments/`), added canonical ICM and RND on the shared CNN encoder, and added a per-step position log. The environment geometry is reused from our own `miniworld.play/envs.py` (a faithful port of the upstream maze). Each run writes a per-update CSV (`update`, `step`, `frames`, `fps`, `visited_count`, `coverage_frac`, `beyond_wall_frac`, `time_at_wall_frac`, `int_rew_mean`, plus model/policy losses) and a compact `.npz` of (`step`, `x`, `z`, `action`, `sticky`). A grid runner launches all runs sequentially (resumable); an analysis script aggregates the CSVs and position logs into the coverage table, the coverage curves, the beyond-wall and time-at-wall figures, and the heatmap-evolution figures. All deviations are logged in `LPM_exploration/UPSTREAM.md`.

7 Limitations (stated up front)

1. **Short budget, small sample.** 2×10^4 steps $\times$ 2 seeds, single environment, yields *qualitative trends*, not rigorous statistical significance; several metrics have between-seed std comparable to their mean. A faithful reproduction needs the upstream 5×10^4 steps with many more seeds (and ideally vectorised environments on a cluster).
2. **Intrinsic signal does not yet dominate.** The no-intrinsic NONE baseline is competitive on coverage, so at this budget exploration is largely driven by policy entropy rather than the intrinsic reward—which is why the coverage ranking (unlike the action-fixation result) does not separate the methods. This is a scale limitation, not evidence against the methods.
3. **Uniform reward normalisation is a design choice.** Running-std normalisation makes a single λ_i comparable across methods but deviates from the upstream per-method offsets; within-method clean-vs-noisy comparisons are the most reliable read.
4. **ICM/RND are our ports.** They are canonical implementations on the shared encoder, not upstream code; the LPM and MSE paths are the *verbatim*-extracted ones.
5. **Coverage is a 2D projection.** Occupancy is binned on the floor plane; agent heading and the first-person view (what actually drives the pixel predictors) are not visualised here.